

# PROJECT CHARTER

## Software Project Management (SPM)

### AI-Based Cloud Detection and Localization in Satellite Images


Project Members:

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# AI-Based Cloud Detection and Localization in Satellite Images

## 1. Project Overview

Project Title
AI-Based Cloud Detection and Localization in Satellite Images

Project Type
Artificial Intelligence / Machine Learning / Image Processing

Primary Platform
Python-based development environment
None for the proposed student version
Hardware Requirement

Core Output
AI model and application that detects and highlights cloud regions in satellite images

## 2. Project Background

The source project proposes an AI-based approach for detecting clouds in optical satellite imagery and optimizing the model for feasible onboard deployment. For this SPM, the onboard-deployment aspect is kept outside the core scope. The student project focuses on cloud detection, localization, visualization, and performance evaluation.

## 3. Problem Statement

Clouds can obscure the Earth's surface in optical satellite imagery, reducing the usefulness of images for observation and analysis. Manually identifying cloud-covered areas is time-consuming when many images must be processed. The proposed system will use AI and image-processing techniques to automatically detect clouds and identify their locations in satellite images.

## 4. Project Objectives

- Collect a suitable labeled satellite-image dataset.
- Preprocess and prepare images for model development.
- Develop and train an AI model for cloud detection.
- Evaluate the model using appropriate performance metrics.
- Localize or highlight detected cloud regions.
- Estimate cloud coverage where feasible.
- Develop a simple interface for uploading an image and viewing results.
- Document the system, experiments, results, limitations, and future work.

## 5. Proposed System

Satellite Image → Preprocessing → AI Model → Cloud Detection → Cloud Localization → Result → Visualization

## 6. Functional Requirements

- The system shall determine whether cloud regions are present.
- The system shall highlight or localize detected cloud regions where the implemented model supports
- localization.
- The system shall display the original and processed images.
- The system shall provide model performance results during evaluation.
- The system shall provide a simple user interface for demonstration.
- The system shall accept a satellite image as input.
- The system shall preprocess the image before inference.

## 7. Non-Functional Requirements

- The application should be understandable and easy to operate.
- The model and results should be reproducible using the documented dataset and procedure.
- The system should provide reasonable processing time for demonstration images.
- The project should maintain clear documentation of datasets, experiments, and limitations.

## 8. Scope

In Scope:

- Dataset preparation
- Image preprocessing
- AI model training and testing
- Cloud detection
- Basic cloud localization
- Cloud-coverage estimation where feasible
- Result visualization
- Performance evaluation
- Simple user interface
- Project documentation

## 9. Suggested Technology Stack

- Programming Language
- Python
- OpenCV / NumPy
- Image Processing
- AI Framework
- TensorFlow or PyTorch
- Jupyter Notebook / Google Colab
- Development
- Visualization
- Matplotlib
- Optional Interface
- Streamlit or Flask

## 10. Major Deliverables

1.Prepared dataset and preprocessing pipeline
2.Trained AI model
3.Cloud detection module
4.Cloud localization/visualization module
5.Performance evaluation report
6.Working demonstration interface
7.Final SPM report and presentation

**11. Milestones**

- ➔ Phase 1:Requirements, literature review, and dataset selection
- ➔ Phase 2:Dataset preparation and preprocessing
- ➔ Phase 3:Baseline model development
- ➔ Phase 4:Training and evaluation
- ➔ Phase 5:Cloud localization and visualization
- ➔ Phase 6:Interface development
- ➔ Phase 7:Testing, documentation, and final demonstration
- ➔

**12. Team Roles**

Member 1:
Dataset collection, labeling/verification, and
preprocessing
Member 2:
AI model development and training
Member 3:
Localization, image processing, and evaluation
Member 4:
Interface, integration, testing, and documentation

**13. Risks and Mitigation**

- Poor dataset quality
- Use a suitable labeled open dataset and verify samples.
- Low model accuracy
- Improve preprocessing, training settings, or model
- selection.
- Training time
- Use Google Colab/GPU resources where available.
- Localization becomes too difficult
- Keep classification as the minimum viable product and
- localization as the next milestone.
- Scope expansion
- Freeze core requirements before implementation and
- treat advanced features as future work.

**14. Success Criteria**

- A working system can accept satellite images.
- The trained model can detect cloud presence with measurable performance.
- Cloud regions can be visualized/localized within the selected project scope.
- The team can demonstrate the system using previously unseen test images.
- The project includes documented experiments, evaluation metrics, limitations, and conclusions.